Making long term infrastructure costs visible

How might we help residents weigh an infrastructure project's upfront price against its future maintenance, operating and environmental costs?

Build a working proof of concept that a municipality could realistically pilot. It can be a web app, data tool, map, dashboard or another format. A municipal leader should be able to use it without your team in the room.

The problem

When a municipality approves a new road, recreation centre or pipe network, public debate usually centres on the upfront capital cost; the price to build a road, arena, pipe, library or new subdivision's servicing. An infrastructure project's initial price is easier to see than its future operating, maintenance and environmental costs. Councils and residents need a clearer picture of the trade-offs over decades. An asset's operating, maintenance, renewal and eventual replacement costs can far exceed the price of building it. Those future costs land on property taxes and utility rates for decades. Environmental costs, such as emissions, energy use and climate risk, are even less visible.

Ontario's municipal asset management framework considers infrastructure across its lifecycle: construction, operation, maintenance, renewal and decommissioning. It connects costs with service levels and risks. A lower initial price can therefore lead to different long-term costs. [1, 2] Many municipalities face large backlogs of ageing infrastructure.

Toronto publishes a ten-year capital plan and utility records for defined City-owned and operated facilities. These can inform an infrastructure comparison, but do not provide a complete lifetime estimate. Future costs and environmental effects require clearly stated evidence and assumptions. [3, 4]

The long-term costs of a new project are rarely presented to residents in a way they can compare or understand. When only the upfront price is visible, decisions can favour options that look cheap now but are expensive over time. Low-density growth, for example, can require more roads and pipes per household to maintain. Senior government grants often cover construction but not upkeep, shifting long-term costs onto local taxpayers.

Tools that show lifecycle costs clearly and fairly could improve public debate and help councils explain why a more expensive option today may cost less overall.

How common and persistent it is

- Canada's core public infrastructure has a replacement value of $2.1 trillion, and municipalities own about 62% of it (Statistics Canada, 2020 data).
- About $265 billion of that infrastructure is in poor or very poor condition, roughly $16,250 per private dwelling. Roads make up nearly half of it. A further $196 billion is in unknown condition.
- Many smaller Ontario municipalities report reinvesting well below their target rate. For example, Markstay-Warren reports an actual reinvestment rate of 0.39% against a 2.77% target (Markstay-Warren's Asset Management Plan, 2024).

Questions to investigate

- How are construction, operating, maintenance and replacement costs accounted for across municipal budgets, and where is the connection between them difficult to see?
- How do grants, debt and other funding sources affect what a project costs residents upfront and over time?
- What long-term infrastructure responsibilities come with new development, and how are those costs considered when growth is approved?
- How can residents and decision-makers weigh costs and benefits that extend across several council terms and generations?
- What can asset management plans reveal about future costs, and how could that information become easier for residents to understand?
- How should comparisons account for environmental impacts and uncertainty about asset lifespan, future prices and maintenance needs?
- Is the publicly available information currently easy to access and understand?

What it costs

- Higher long-term spending as deferred maintenance turns into more expensive full replacements. Huntsville's asset plan estimates proactive lifecycle management saves 45% annually on roads (Town of Huntsville's 2024 Asset Management Plan).
- Service risks such as watermain breaks, road closures and facility shutdowns when assets fail.
- Intergenerational fairness when today's decisions shift costs to future taxpayers.
- Trust can be eroded when residents feel blindsided by tax and rate increases they didn't see coming.

Who you are designing for

Primary users. Residents and ratepayers trying to understand infrastructure spending and long-term value.

Municipal users. Councillors, finance staff, asset managers and operations teams comparing investment options and making budget and capital-plan decisions.

Other stakeholders. Facility and service users, future residents and communities affected by construction, emissions or service changes.

What success looks like

- Users can compare upfront spending with operating, maintenance, renewal and end-of-life costs over a stated period.
- The tool makes service and environmental trade-offs understandable alongside financial costs.
- Users can change important assumptions and see how the comparison changes.
- A municipal leader can use the working tool independently and understand which conclusions are supported and which need further evidence.
- Lifecycle cost modelling: estimating costs over 25 to 100 years requires assumptions about asset life, maintenance cycles, inflation and discount rates

- Discounting as the choice of discount rate strongly shapes results; tools should make it visible and adjustable
- Showing costs over time, per household or per year so non-experts can compare options.
- Assumptions that are visible, explained and adjustable.

Constraints

- Compare options that provide a clearly stated service. Identify the asset, study period, price year and costs included.
- Keep capital, operating and renewal costs separate and avoid double counting. Explain inflation or discounting if used.
- Show sourced values, estimates and assumptions clearly. Test uncertainty in major inputs such as energy prices, maintenance and asset life.
- Keep environmental measures understandable and consistent. Operational emissions do not capture every construction impact; explain any monetary value assigned to them.

Public datasets to explore

You may use other public data that better supports your solution. Record your sources and dates, and make data limitations visible.

1. **Capital Budget and Plan by Ward (City of Toronto data and downloads)**

Includes project and subproject names, annual allocations and ten-year totals. Amounts are in thousands of dollars. Use relevant capital spending as a starting point; a ten-year allocation is not necessarily the full project cost or actual spending.

2. **Annual Energy Consumption City Owned and Operated Buildings (City of Toronto data and downloads)**

This file has property and utility meter tables linked by Portfolio Manager ID, with consumption, units, dates and costs where populated. It can support a facility operating baseline. Missing costs are not zero, and this file has no direct greenhouse gas totals.

3. **Statistics Canada, Canada′s Core Public Infrastructure Survey tables.**

Including Table 34-10-0284-01 (estimated replacement value by physical condition rating) and Table 34-10-0289-01 (inventory by year of construction). See the table list in The Daily, October 2024. Useful for realistic replacement values, condition mix and asset age by type.

4. **Ontario Financial Information Return (FIR) data** (Ontario Ministry of Municipal Affairs and Housing; XLSX/CSV, with newer years on the FIR open data site).

Includes municipal revenues, capital and operating expenses, reserves and tangible capital assets for every Ontario municipality.

Other potential sources are municipal asset management plans (required in Ontario; most are public PDFs with replacement costs, condition and lifecycle needs); capital and operating budgets; Ontario Financial Information Returns (FIR); Statistics Canada′s Core Public Infrastructure Survey; Canadian Infrastructure Report Card.

Explain your thinking

Show who the tool serves, why you designed it this way and how you checked its outputs. Include clear instructions, source links and any assumptions. Explain what a municipality would need to test or add before piloting it. If you use AI, make its role and limitations clear.

Background sources

[1] Government of Ontario. Municipal asset management planning

[2] Government of Ontario. Asset Management Planning for Municipal Infrastructure

[3] City of Toronto. Capital Budget and Plan by Ward

[4] City of Toronto. Annual Energy Consumption